Lecture 2: Planet Formation 101 & Orbital Dynamics

Introduction to Planetary Science

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Protoplanetary disk HL Tauri. Credit: ALMA/ESO/NAOJ/NRAO

Learning Objectives

By the end of this lecture, you will be able to:

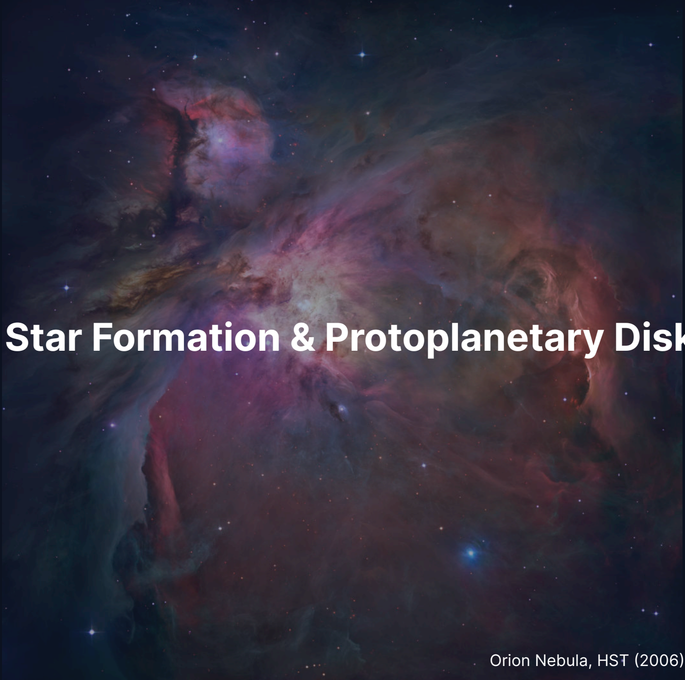
- Describe the stages of planet formation from molecular cloud to planets
- Explain how dust grows into planetesimals and planets
- Apply Kepler's laws and the vis-viva equation to orbital problems
- Discuss orbital resonances, tidal forces, and planetary migration

Recap: Lecture 1

- Three driving questions: formation, habitability, life
- The IAU planet definition and the Pluto debate
- Solar system architecture: inner rocky, outer gas/ice
- Derived $M_\odot$ from Kepler's third law
- Comparative planetology as methodology

Today: How did all of this come to be?

1. Star Formation & Protoplanetary Disks



Orion Nebula, HST (2006). Credit: NASA/ESA/M. Robberto

From Molecular Cloud to Star

- Stars form from cold (T ~ 10–20 K), dense molecular clouds
- Collapse triggered when self-gravity overcomes thermal pressure
- Critical mass for collapse, the **Jeans mass**:

$$M_J \sim \frac{c_s^3}{\sqrt{G^3 \rho}}$$

- For typical conditions: $M_J \sim 1\text{–}10\ M_\odot$
- Angular momentum conservation ⇒ **disk** forms around the protostar

Jeans Instability: When Gravity Wins

Derivation. Compare gravitational free-fall time to sound-crossing time.

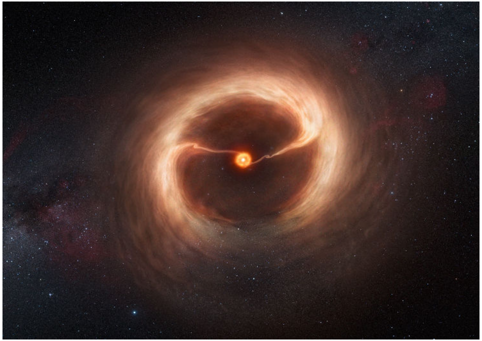
- Free-fall time: $t_{\mathrm{ff}} = \sqrt{3\pi/(32\,G\rho)}$
- Sound-crossing time across scale L: $t_{\mathrm{sc}} = L/c_s$
- Collapse requires $t_{\mathrm{ff}} < t_{\mathrm{sc}}$, giving:

$$L > L_J = c_s\, t_{\mathrm{ff}} \sim \frac{c_s}{\sqrt{G\rho}}$$

- For typical cloud conditions (T = 10 K, $n_{\mathrm{H}_2} = 10^4\ \mathrm{cm}^{-3}$): $L_J \sim 0.1$ pc, $M_J \sim 1\text{–}10\,M_\odot$
- Free-fall time ~ 10^5 yr
- Fragmentation: if the cloud contains multiple Jeans masses, it can split into many protostars

Source: Jeans (1902); Shu (1977)

Protoplanetary Disk Structure



- Flattened rotating structure, ~few AU to ~hundreds AU
- Total mass ~0.01–0.1 $M_\odot$
- **Temperature gradient:** >1500 K (inner) to <30 K (outer)
- **Dust-to-gas ratio:** ~1:100 (by mass)
- Surface density: $\Sigma(r) \propto r^{-p}$

Vertical Structure and Flaring

The disk is not razor-thin: it has a finite vertical scale.

- Hydrostatic balance gives a Gaussian density profile with scale height H
- $H/r = c_s/v_{\mathrm{K}}$ where c_s is sound speed and v_{K} is Keplerian speed
- Typical $H/r \sim 0.03$–0.1, increases outward (flared disk)
- Dust **settles** toward the midplane under vertical gravity
- Settling creates a denser dust layer ($H_{\mathrm{dust}}/H_{\mathrm{gas}} \sim 0.1$)
- Once the midplane dust-to-gas ratio approaches unity, instabilities take over

Dust settling is the first step toward planetesimal formation: it creates the conditions where streaming instability can act.

The Minimum-Mass Solar Nebula

If we spread the solar system's planets back into a disk, what mass would we need?

- Augment each planet's mass by solar H/He to match solar composition
- Spread over an annulus centred on each planet
- Result: **Minimum-Mass Solar Nebula (MMSN)**, Hayashi (1981)
- Total mass: ~0.01–0.02 $M_\odot$
- Surface density: $\Sigma_{\mathrm{MMSN}}(r) \approx 1700\,(r/\mathrm{AU})^{-3/2}$ g/cm^2
- A convenient scaling, but modern observations favour more massive disks for Jovian cores

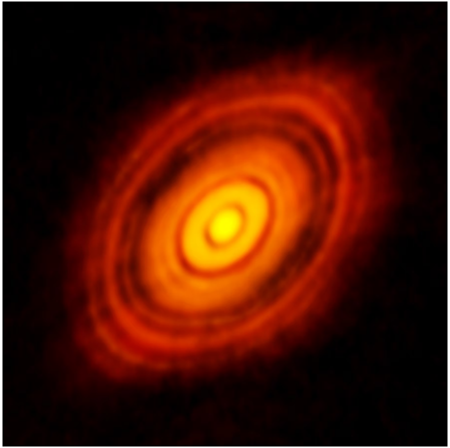
Source: Hayashi (1981); Desch (2007) for revisions

The Snow Line

- **Snow line** = distance where water condenses as ice (~150–170 K)
- In our solar system: ~2.5–3.5 AU (between Mars and Jupiter)
- Beyond the snow line:
  - Solid material ~3× more abundant (ice added to rock/metal)
  - Enables faster growth of planetary cores
- Other ice lines at greater distances: CO_2, CO, N_2, NH_3
- Creates a **compositional gradient** across the disk

The snow line explains the fundamental architecture of the solar system: small rocky planets inside, massive gas/ice giants outside.

ALMA: Disks Resolved

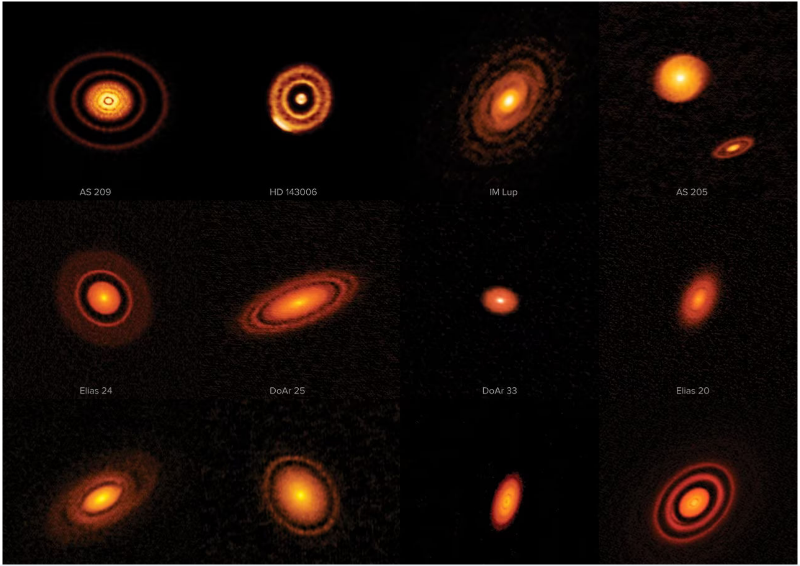


Credit: ALMA/ESO/NAOJ/NRAO (2014)

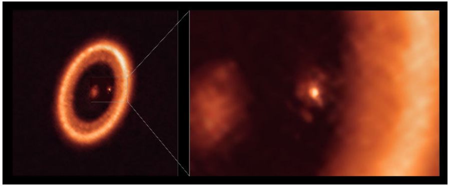
- **HL Tauri** (2014): ~1 Myr old star
- Concentric rings and gaps at AU resolution
- Gaps possibly carved by forming planets
- Disk substructure present even in very young systems

Planet formation begins earlier than previously thought.

DSHARP Survey: Substructure Is Universal



PDS 70: Planets Caught in the Act



Credit: ALMA/ESO/NAOJ/NRAO (Benisty et al. 2021)

In 2018–2019, VLT/SPHERE and ALMA directly imaged two protoplanets in the disk gap.

- PDS 70 b: ~6 M_{Jup}, ~20 AU
- PDS 70 c: similar mass, at ~35 AU
- Hα emission: active accretion onto both planets
- Circumplanetary disk around PDS 70 c (2021)
- First direct evidence of giant planets forming in disk gaps
- Supports the DSHARP picture: gaps sculpted by embedded protoplanets

ALMA MAPS: Chemical Diversity in Disks

The Molecules with ALMA at Planet-Forming Scales (MAPS) survey (2021) resolved the chemical composition of 5 disks.

- Mapped the distributions of CO, HCO^+, HCN, C_2H, CH_3CN, H_2CO, ...
- Showed that complex organic molecules are present at planet-forming distances
- Different chemistries at different radii: volatile C/O ratios vary with distance
- Snow lines for multiple species are directly observable
- Implications: the compositions inherited by forming planets depend on where in the disk they assemble

Source: Öberg et al. (2021), *ApJS*

Disk Lifetimes: A Hard Deadline

- Infrared surveys of young stellar clusters show disk dispersal:
  - ~80% of stars have disks at 1 Myr
  - <10% by 10 Myr
- **Median disk lifetime: 3–5 Myr**
- Dispersal driven by:
  - Viscous accretion onto the star
  - Photoevaporation by stellar UV/X-ray radiation
  - Magnetohydrodynamic winds

The disk is gone by ~3–5 Myr. Giant-planet **cores** must reach critical mass well before that, typically in the first 1–3 Myr, to capture substantial gas envelopes (Lambrechts & Johansen 2014).

2. From Dust to Planets

HL Tauri, ALMA (2014). Credit: ALMA/ESO/NAOJ/NRAO

Grain Growth: Sticking and Settling

- Starting material: sub-μm dust grains (silicates, oxides, ices)
- Growth factor to planets: ~10^{13} in size
- **Step 1: Sticking**
  - Low-velocity collisions ($\lesssim 1\,\mathrm{m\,s^{-1}}$): van der Waals forces
  - Forms fractal aggregates up to mm–cm sizes
- **Step 2: Settling**
  - Dust settles to the disk midplane under vertical gravity
  - Creates a dense dust layer within the gas disk

Dust–Gas Coupling: The Stokes Number

How tightly is a particle coupled to the gas?

$$\mathrm{St} = \tau_{\mathrm{stop}}\,\Omega_K$$

where τ_{stop} is the friction stopping time and Ω_K is the Keplerian frequency.

$\mathrm{St} \ll 1$ (small grains) perfectly coupled to gas; follow the flow

$\mathrm{St} \sim 1$ (pebbles, cm-dm) partially decoupled; experience maximum radial drift

$\mathrm{St} \gg 1$ (boulders, km) decoupled; move on Keplerian orbits

Particles with $\mathrm{St} \sim 0.1$ are the “sweet spot” for pebble accretion: they drift efficiently yet can still be deflected by planetary gravity.

Source: Weidenschilling (1977); Johansen et al. (2014) in *Protostars & Planets VI*

Growth Barriers

Three barriers impede growth beyond cm-scale:

Bouncing barrier (~mm): Compacted aggregates bounce rather than stick

Fragmentation barrier (~cm): Collision velocities >1–10 m s^{-1} shatter aggregates

Radial drift barrier (~metre): Gas headwind drains angular momentum
→ Metre-sized boulders spiral into the star in ~100–1000 yr

The metre-size problem

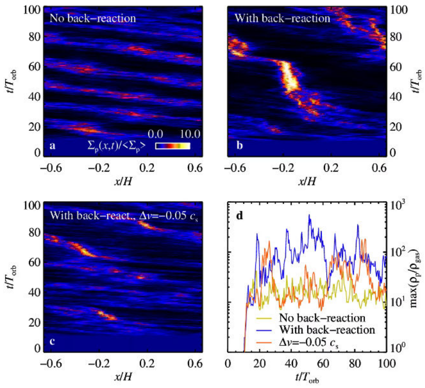
How does nature bridge the gap from cm to km?

Streaming Instability: Skipping the Barrier

- When local dust-to-gas ratio $\gtrsim 1$ (from settling + pressure bumps):
  1. Concentrated particles accelerate gas toward Keplerian speed
  2. Reduced headwind attracts more drifting particles
  3. **Positive feedback** → dense filaments and clumps
- Dense clumps collapse under self-gravity
- Directly form **planetesimals** (~50–500 km)
- Skips the problematic cm-to-metre range entirely

The streaming instability solves the radial drift barrier by collective effects: particles help each other avoid falling into the star.

Streaming Instability: A Simulation Snapshot



Credit: Johansen et al. (2007), Nature 448, 1022 (arXiv:0708.3890)

- **Panels a–c**: particle column density vs. radial position and time
- **Panel a**: no back-reaction on gas → only weak drift banding
- **Panel b**: with back-reaction → dense clumps form
- **Panel d**: maximum particle density rises to >10^3 background
- Clumps are dense enough for self-gravitating

Pebble Accretion: Fast Growth

- Once planetesimals exist, they sweep up mm–cm "pebbles"
- Gas drag deflects pebbles strongly toward growing body
- Effective cross-section $\gg$ geometric cross-section
- Enables growth to ~10 $M_\oplus$ within disk lifetime
- Fast enough to trigger gas accretion → giant planets

Planet formation sequence

Dust grains $\xrightarrow{\text{sticking}}$ aggregates $\xrightarrow{\text{streaming inst.}}$ planetesimals $\xrightarrow{\text{pebble acc.}}$ planetary cores

Gravitational Focusing

Why does the largest body in a swarm grow fastest?

$$\sigma_{\mathrm{eff}} = \pi R^2 \left(1 + \frac{v_{\mathrm{esc}}^2}{v_\infty^2}\right)$$

- Classical hard-sphere cross-section = πR^2
- Gravity of the body curves nearby trajectories inward
- Effective collision cross-section is enhanced by $(1 + v_{\mathrm{esc}}^2 / v_\infty^2)$
- When $v_{\mathrm{esc}} \gg v_\infty$ (low random velocities), focusing is enormous
- Because $v_{\mathrm{esc}} \propto \sqrt{M/R}$, the **largest** body focuses most effectively

Consequence: an initially small mass excess runs away into a dominant body.

Runaway & Oligarchic Growth

Runaway growth:

- Biggest body focuses best
- Grows fastest (positive feedback)
- Reaches Moon–Mars mass in ~10^5 yr
- Ends when the big body stirs up the swarm

Oligarchic growth:

- Winners stir up leftovers → focusing weakens
- A few dozen "oligarchs" dominate
- Each separated by ~5–10 Hill radii
- Isolation mass: ~0.01–0.1 $M_\oplus$

Isolation Mass

How big can oligarchs get?

The **isolation mass** is reached when an oligarch has consumed all available planetesimals within its feeding zone:

$$M_{\mathrm{iso}} \sim \frac{(4\pi B a^2 \Sigma_s)^{3/2}}{(3M_*)^{1/2}}$$

where $B \sim 5$–10 is the feeding zone width in Hill radii.

- Inner solar system ($\Sigma_s \sim 10$ g/cm^2): $M_{\mathrm{iso}} \sim 0.01$–$0.1\ M_\oplus$
- Outer solar system (beyond snow line, $\Sigma_s \sim 30$ g/cm^2): $M_{\mathrm{iso}} \sim 10\ M_\oplus$
- Explains why giant planet cores can reach critical mass outside the snow line but not inside

Core Accretion: Building Giant Planets

1. Solid core grows beyond the snow line (more solids available)
2. Core reaches **critical mass** ~5–15 $M_\oplus$
3. Gravity binds a gas envelope from the disk
4. Envelope contracts, cools → **runaway gas accretion**
5. Accretion stops when planet opens a gap or disk disperses

This explains:

- Jupiter & Saturn: formed beyond snow line, reached critical mass early
- Uranus & Neptune: reached critical mass too late, less gas captured

Alternative: Gravitational Instability

- If the disk is massive and cools rapidly:
- Gas disk fragments directly into giant planets
- Very fast: ~10^3 yr (vs. ~10^6 for core accretion)
- May work at large orbital distances (≳30–50 AU)
- Debated for solar system giants

Two pathways

Core accretion:
bottom-up (dust → core → gas)

Grav. instability:
top-down (disk → clump → planet)

Planet–Planet Scattering

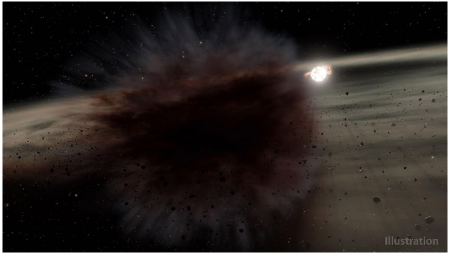
After the gas disk disperses, gravitational interactions among embryos become **chaotic**.

- Initially packed oligarchs cross each other's Hill radii
- Close encounters pump eccentricities on ~Myr timescales
- Orbits intersect ⇒ collisions or ejections
- Survivors spiral toward stable separations (~10–20 mutual Hill radii)
- Damping by residual planetesimals circularises orbits over 10–100 Myr

Scattering is the dominant mechanism for producing the observed eccentricity distribution of exoplanets, including eccentric giant planets at intermediate separations.

Source: Rasio & Ford (1996); Chambers (2004); Lissauer et al. (2011)

The Giant Impact Phase

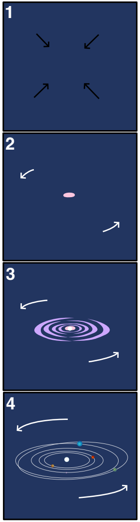


Credit: NASA/JPL-Caltech (artist's concept)

- After disk dispersal: embryos' orbits become chaotic
- ~10–100 Myr of collisions assemble final terrestrial planets
- **Moon-forming impact:** proto-Earth hit by Mars-sized Theia
- Melts the planet → magma ocean → core formation

Formation is not just assembly; it drives differentiation, outgassing, and sets the stage for all subsequent evolution (Lectures 3 & 4).

From Dust to Planets: The Full Sequence



Credit: Wikimedia, public

Four key stages, from top to bottom:

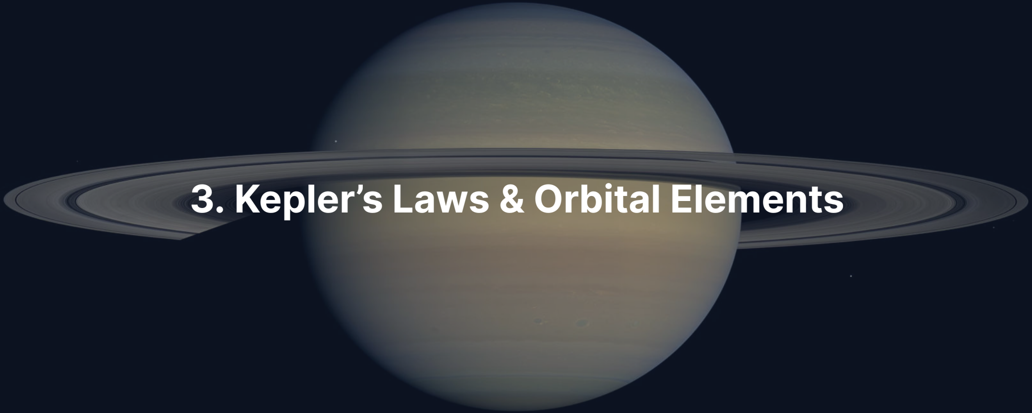
1. **Cloud collapse:** a cold, dense molecular cloud fragments and contracts under self-gravity
2. **Protoplanetary disk:** conservation of angular momentum flattens the infalling gas into a rotating disk
3. **Disk substructure:** dust settles and concentrates into rings; streaming instability builds planetesimals
4. **Mature system:** runaway and oligarchic growth, followed by giant impacts, produce the planets on stable orbits

Total timescale: ~100 Myr from first collapse to a settled planetary system.

Break

Earthrise, Apollo 8 (1968). Credit: NASA

3. Kepler’s Laws & Orbital Elements



Saturn’s rings, Cassini (2009). Credit: NASA/JPL/SSI

Kepler's Three Laws

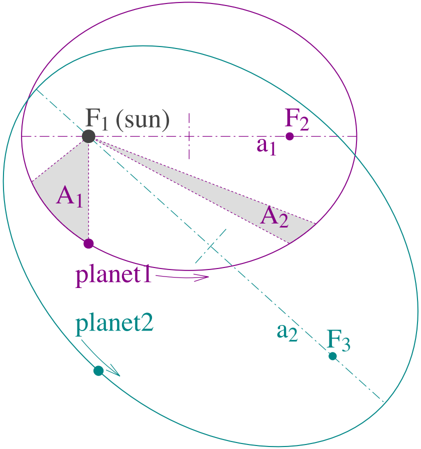
1. **Law of ellipses:** Each planet orbits the Sun on an **ellipse** with the Sun at one focus.
2. **Equal areas:** A line from planet to Sun sweeps **equal areas in equal times**.
   $\Rightarrow$ faster at perihelion, slower at aphelion
3. **Harmonic law:**

$$P^2 = \frac{4\pi^2}{G(M_1 + M_2)} a^3$$

   Connects period P to semi-major axis a and total mass.

Newton showed: **all three follow from universal gravitation**.

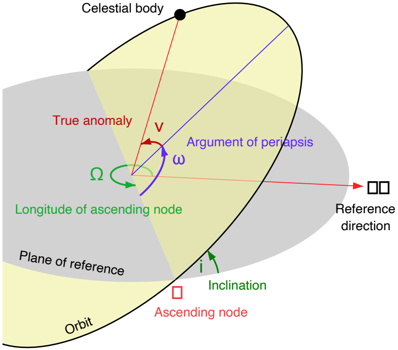
Visualising Kepler’s Laws



Credit: Hankwang, CC BY 2.5

Elliptical orbit (left), equal areas (centre, shaded regions have equal area), and the period–semi-major axis relation (right).

Six Orbital Elements



Credit: Lasunncty, CC BY-SA 3.0

Element	Symbol	Controls
Semi-major axis	a	Size
Eccentricity	e	Shape
Inclination	i	Tilt
Long. asc. node	Ω	Twist
Arg. perihelion	ω	Orientation
True anomaly	ν	Position

First five = orbit shape & orientation.
Sixth = where the body is now.

The Two-Body Problem

Two masses m_1, m_2 interacting through gravity:

- Reduces to **reduced mass** $\mu = m_1 m_2/(m_1 + m_2)$ in central field of $M = m_1 + m_2$
- Total orbital energy:

$$E = \frac{1}{2}\mu v^2 - \frac{Gm_1 m_2}{r}$$

- Fundamental result: energy depends **only on** a:

$$E = -\frac{Gm_1 m_2}{2a}$$

Same a → same energy, regardless of eccentricity.

4. Blackboard Derivation: Vis-Viva Equation

Jupiter, Hubble (2014). Credit: NASA/ESA/A. Simon

Deriving the Vis-Viva Equation

Goal: Find the speed v at any distance r in an orbit with semi-major axis a.

Starting point: Total energy is conserved and equals $E = -GMm/(2a)$.

At any point on the orbit:

$$\frac{1}{2}mv^2 - \frac{GMm}{r} = -\frac{GMm}{2a}$$

where M is the central mass ($m \ll M$).

Step 1: Evaluate at Perihelion & Aphelion

At perihelion ($r_p = a(1 - e)$) and aphelion ($r_a = a(1 + e)$):

$$\frac{1}{2}mv_p^2 - \frac{GMm}{a(1-e)} = \frac{1}{2}mv_a^2 - \frac{GMm}{a(1+e)}$$

Angular momentum conservation:

$$v_p \cdot a(1-e) = v_a \cdot a(1+e)$$

$\Rightarrow v_p/v_a = (1+e)/(1-e)$

Solving yields $E = -GMm/(2a)$

Key insight: Energy depends only on a, **not** on e.

Step 2: The Vis-Viva Equation

Setting instantaneous energy equal to the total:

$$\frac{1}{2}mv^2 - \frac{GMm}{r} = -\frac{GMm}{2a}$$

Dividing by $m/2$ and rearranging:

$$\boxed{v^2 = GM\left(\frac{2}{r} - \frac{1}{a}\right)}$$

Important special cases:

- **Circular orbit** ($r = a$): $v_{\text{circ}} = \sqrt{GM/a}$
- **Escape velocity** ($a \to \infty$): $v_{\text{esc}} = \sqrt{2GM/r} = \sqrt{2}\, v_{\text{circ}}$

Application: Earth's Orbital Velocity

Point	Distance r	Velocity
Perihelion	$a(1-e) = 1.471 \times 10^{11}$ m	$v_p = 30.3\ \mathrm{km\,s^{-1}}$
Aphelion	$a(1+e) = 1.521 \times 10^{11}$ m	$v_a = 29.3\ \mathrm{km\,s^{-1}}$
Mean	$a = 1.496 \times 10^{11}$ m	$\bar{v} = 29.8\ \mathrm{km\,s^{-1}}$

Nearly circular orbit ($e = 0.017$) → only $\pm 0.5\ \mathrm{km\,s^{-1}}$ variation.

Application: Halley's Comet

Halley: a = 17.8 AU, e = 0.967 (extreme eccentricity)

Point	**Distance**	**Velocity**
Perihelion	0.586 AU	54.6 $\mathrm{km\,s^{-1}}$
Aphelion	35.1 AU	0.91 $\mathrm{km\,s^{-1}}$

- Races through the inner solar system at ~2× Earth's speed
- Crawls beyond Neptune at <1 $\mathrm{km\,s^{-1}}$
- Same total energy, just redistributed between K and U along the orbit

5. Orbital Resonances & Tidal Forces

Jupiter, Hubble (2014). Credit: NASA/ESA/A. Simon

Mean-Motion Resonances

- *p*:*q* **resonance**: inner body completes *p* orbits per *q* of the outer body
- Gravitational kicks add up coherently → can stabilise **or** destabilise

Resonance	Bodies	Effect
3:2	Pluto–Neptune	Protects Pluto from close encounters
Near 5:2	Jupiter–Saturn	“Great inequality”
2:1	Asteroid belt gaps	Kirkwood gaps (destabilising)
1:1	Jupiter Trojans	Co-orbital at L4, L5

The Laplace Resonance

The Galilean moons Io, Europa, Ganymede are in a **1:2:4 period ratio**:

Moon	Period (days)	Ratio to Io
Io	1.769	1
Europa	3.551	≈2
Ganymede	7.155	≈4

- Maintained by tidal interactions with Jupiter
- **Forces non-zero eccentricities**
- Io's $e \approx 0.004$ drives tidal flexing → extreme volcanism (Lecture 3)

Orbital resonances can drive geological activity billions of years after formation.

Secular Resonances: Slow Dynamical Chaos

Beyond mean-motion resonances, orbits are also coupled by **secular** (long-term) precession frequencies.

- A secular resonance occurs when the apsidal or nodal precession frequency of one body matches another's
- In the solar system: v_5 (Jupiter's apsidal frequency) and v_6 (Saturn's) resonances define the inner edges of the asteroid belt
- Asteroids near v_6 at 2.1 AU have their eccentricity pumped until they are ejected or collide with the Sun
- Secular chaos operates on 10^4–10^7 yr timescales
- Laskar (1989): Earth's eccentricity evolution is **intrinsically chaotic** beyond ~60 Myr
- Relevance: explains Kirkwood gaps and long-term orbital instabilities

Source: Laskar (1989, 1990); Murray & Dermott (1999)

Tidal Forces

- Gravity varies across an extended body → **differential force**
- Tidal acceleration across a moon of radius R_s at distance d:

$$\Delta a_{\mathrm{tidal}} \approx \frac{2GM_pR_s}{d^3}$$

- Falls off as d^{-3} (steeper than gravity's d^{-2})
- **Tidal bulges:** elongation along the line connecting the bodies
- If rotation ≠ orbital period → bulge misalignment → torque
- Energy dissipated as heat → **tidal heating** (Lecture 3)

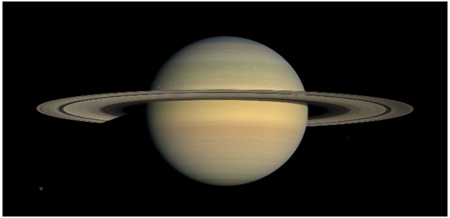
Tidal Locking (Synchronous Rotation)

- Tidal torque gradually slows rotation until $P_{\mathrm{rot}} = P_{\mathrm{orb}}$
- Same face always points toward the planet
- The Moon is tidally locked to Earth (same face always visible)
- All four Galilean moons are tidally locked to Jupiter
- Locking timescale: $\tau_{\mathrm{lock}} \propto d^6/(M_p^2 R_s^3)$

Note

Tidal locking is the **equilibrium state**. Tidal heating occurs during the approach to equilibrium (or when resonances maintain eccentricity, preventing full equilibrium).

The Roche Limit



Credit: NASA/JPL/SSI (Cassini)

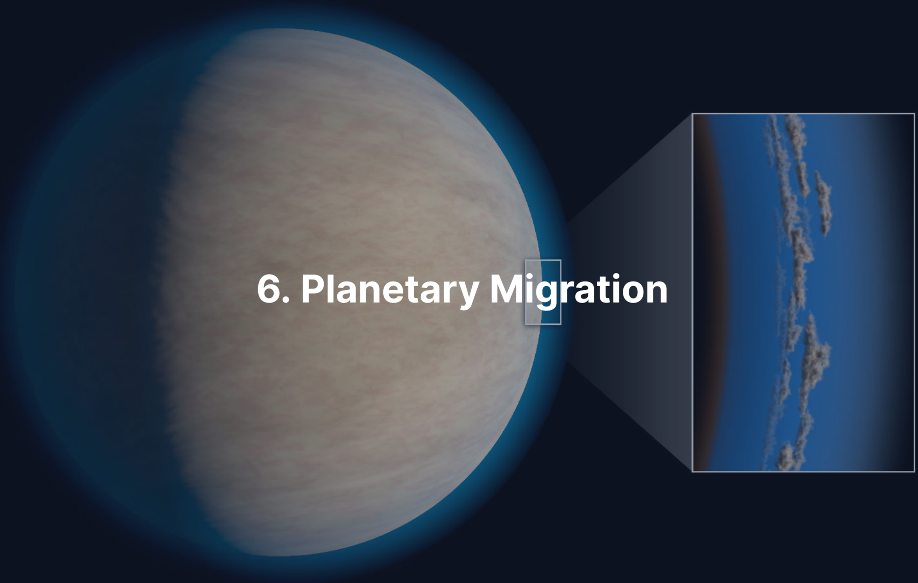
Inside the **Roche limit**, tidal forces exceed self-gravity:
→ satellites are torn apart

For a fluid satellite:

$$d_R \approx 2.46\, R_p \left(\frac{\rho_p}{\rho_s}\right)^{1/3}$$

For Saturn + icy satellite:
$d_R \approx 126{,}000$ km $\approx 2.16\, R_{\mathrm{Saturn}}$
Saturn's main rings extend from 67,000 to 137,000 km, **mostly within the Roche limit** (the formula is approximate; rigid-body and compositional effects shift the boundary).

6. Planetary Migration



Hot Jupiter (artist's concept). Credit: ESO/L. Calçada

Planet–Disk Interactions

- Planets embedded in a disk excite spiral density waves
- Waves carry angular momentum away from the planet
- Net torque can push the planet **inward or outward**

Type I: Low-mass planet ($\lesssim$ few $M_\oplus$), fully embedded
Migrates on ~10^5 yr timescale (potentially too fast!)

Type II: Massive planet (~ M_J) opens a **gap**
Migrates with the disk on viscous timescale (~10^5–10^6 yr)

Type III: Rapid runaway migration in massive disks

Lindblad and Corotation Torques

Two kinds of torques act on a planet embedded in a disk:

- **Lindblad torques:** arise at inner and outer Lindblad resonances where spiral density waves form
  - Inner LR exerts a positive torque (outward push)
  - Outer LR exerts a negative torque (inward push)
  - Outer typically wins slightly $\Rightarrow$ inward migration
- **Corotation torques:** arise from material co-orbiting with the planet
  - Depends sensitively on the disk temperature and density gradients
  - Can be positive (outward) and partially balance Lindblad

The real net torque depends on disk structure; this is an active area of research.

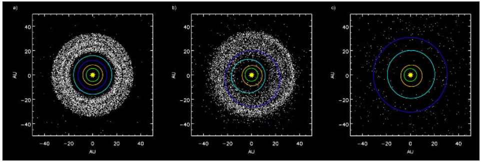
How Does the Gas Disk Go Away?

Planet formation has a hard deadline.

- **Viscous accretion:** gas accretes onto the central star, feeding its growth
- **Photoevaporation:** X-ray and EUV photons from the young star heat the upper disk layers, driving thermal winds
- **MHD winds:** magnetised winds launched from the disk surface carry mass and angular momentum
- Once surface density drops below a threshold, photoevaporation "drains" the remaining gas rapidly
- Total lifetime: 3–5 Myr median (Haisch 2001)

Giant planets must form within 1–3 Myr to capture substantial gas envelopes before the disk disappears.

The Nice Model



Credit: Wikimedia Commons, CC BY-SA 3.0

- Giant planets formed in a **compact** configuration (5–17 AU)
- Jupiter–Saturn crossed 2:1 resonance → **instability**
- Uranus & Neptune scattered outward
- Disrupted the primordial Kuiper Belt
- May explain the Late Heavy Bombardment

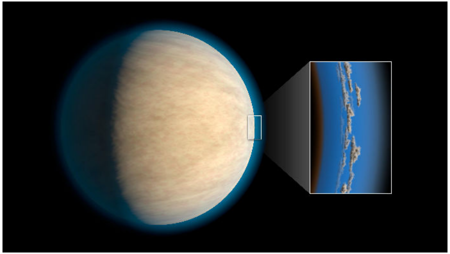
The Grand Tack Hypothesis

1. Jupiter migrated **inward** to ~1.5 AU (Type II migration)
2. Saturn caught up, trapped in resonance
3. Combined torques drove both planets **outward** (“tacking”)
4. Jupiter swept through the inner solar system twice

Potentially explains:

- The small mass of Mars
- The compositional structure of the asteroid belt
- The distribution of water-bearing material in the inner solar system

Evidence for Migration: Hot Jupiters



Credit: ESO/L. Calçada

- Gas giants orbiting at $a \lesssim 0.1$ AU
- Orbital periods of just a few **days**
- Cannot have formed in situ (too hot, too little material)
- Must have formed further out and **migrated inward**

Hot Jupiters are the strongest evidence that planetary migration is real.

7. Recent Advances & Outlook

Protoplanetary disks, DSHARP. Credit: ALMA/ESO/NAOJ/NRAO

High-Eccentricity Migration

Not all migration is disk-driven.

- **Planet–planet scattering:** close encounters between giant planets scatter one inward on a highly eccentric orbit
- **Kozai–Lidov mechanism:** an inclined ($i \gtrsim 40°$) outer companion drives secular oscillations that conserve $L_z = \sqrt{1 - e^2}\cos i$, trading inclination for eccentricity (and back)
- Once eccentric, tidal interactions with the star circularise the orbit at short periods
- Produces hot Jupiters on **high-obliquity** orbits (misaligned with stellar spin)
- Observed: ~1/3 of hot Jupiters have spin-orbit misalignments >30°

Source: Rasio & Ford (1996); Winn et al. (2010); Dawson & Johnson (2018)

Meteoritic Dichotomy: NC vs. CC

Meteorites divide cleanly into two isotopic groups:

- **Non-carbonaceous (NC):** ordinary, enstatite, and aubrite chondrites; most iron meteorites; originated in the inner solar system
- **Carbonaceous (CC):** CI, CM, CV, CR, CB chondrites; some iron meteorites; originated in the outer solar system
- Distinct $^{50}\mathrm{Ti}$, $^{54}\mathrm{Cr}$, $^{48}\mathrm{Ca}$, $^{94}\mathrm{Mo}$ isotopic signatures
- Implies two separate reservoirs that did not mix for ~3–4 Myr
- Natural explanation: **Jupiter** formed early, blocking inward drift of pebbles

Source: Kruijer et al. (2017), *PNAS*; Warren (2011)

Sample-Return Missions: Accretion Fossils

Recent missions have delivered pristine asteroid samples to Earth.

- **Hayabusa2** (JAXA): returned 5.4 g of asteroid **Ryugu** in 2020
- C-type rubble-pile asteroid, carbonaceous chondrite affinity
- Contains organic molecules including all RNA/DNA bases
- **OSIRIS-REx** (NASA): returned ~70 g of asteroid **Bennu** in 2023
- Hydrated clays, carbonates, phosphates: aqueous alteration
- Both targets formed in the outer main belt; provide snapshots of primordial accretion

Source: Yada et al. (2022) on Ryugu; Lauretta et al. (2024) on Bennu

Juno: Jupiter's Dilute Core

Juno's gravity measurements (Wahl et al. 2017) revolutionised our picture of Jupiter's interior.

- Pre-Juno assumption: compact heavy-element core, pure H/He envelope
- Juno data: the "core" extends to **30–50%** of Jupiter's radius
- Smooth compositional gradient rather than a sharp boundary
- Total heavy-element content: ~20–40 $M_\oplus$
- Implication: Jupiter's formation was not a clean core-accretion event
- Possible cause: an early giant impact dispersed the compact core (Liu et al. 2019), or gradual mixing during gas accretion

Source: Wahl et al. (2017); Militzer et al. (2022)

More Recent Advances

- **ALMA MAPS (2021):** chemical maps of 5 disks at planet-forming scales
- **JWST protoplanetary disks:** MIRI detects water and organic molecules in the terrestrial zone of young disks (2023–2024)
- **Pebble accretion:** refined timing constraints on giant-planet formation (<1–3 Myr)
- **Meteorite isotopics:** NC-CC dichotomy + Mo isotopes timing of inner vs. outer disk
- **Psyche:** NASA mission (launched October 2023, arrival 2029) to study a metal-rich asteroid, a possible naked core

JWST: Water and Molecules in Terrestrial Zones

JWST's MIRI instrument can detect molecular emission lines from the inner disks of young stars.

- Detected H_2O, CO_2, C_2H_2, HCN, CH_3 in the inner 1–10 AU of several disks
- Water vapour abundance traces ice transport across the snow line
- Pebble drift may enhance inner-disk water content during planet formation
- Evidence that delivery of water to the terrestrial zone is an active, ongoing process during planet formation

Source: Kamp et al. (2023); Banzatti et al. (2023); Perotti et al. (2023)

What Kind of Planets Form?

The outcomes of planet formation, seen in exoplanet surveys:

- **Hot Jupiters:** rare (~1% of solar-type stars), clear evidence for migration
- **Warm/cool gas giants:** ~10% of stars at 1–10 AU
- **Super-Earths / mini-Neptunes:** extremely common, ~30–50% of stars
- **Radius valley:** gap in planet radii around 1.5–2 $R_\oplus$ (Lecture 13)
- **Compact multi-planet systems:** many stars host 3–8 planets inside 1 AU
- **Solar-system analogues:** relatively rare

We will return to exoplanet demographics in Lecture 13.

Summary

1. Planets form in **protoplanetary disks** around young stars (lifetime ~3–5 Myr)
2. Dust grows to planetesimals via sticking, streaming instability, and pebble accretion
3. Giant planets form by **core accretion**: solid core → critical mass → runaway gas capture
4. Orbits follow Kepler's laws; the **vis-viva equation** gives speed at any point
5. Orbital **resonances** can protect or destabilise orbits, and drive tidal heating
6. Planets **migrate** through disk and planetesimal interactions, confirmed by hot Jupiters

Questions?

Next: **Lecture 3. Planetary Heat & Energy Transport**

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`